

DFSMSHsm Fast Replication Part 1

Glenn Wilcock
wilcock@us.ibm.com

August 17, 2006
Session 3038

Legal Disclaimer



- Copyright © 2006 by International Business Machines Corporation.
- No part of this document may be reproduced or transmitted in any form without written permission from IBM Corporation.
- Product data has been reviewed for accuracy as of the date of initial publication. Product data is subject to change without notice. This information could include technical inaccuracies or typographical errors. IBM may make improvements and/or changes in the product(s) and/or program(s) at any time without notice.
- Any statements regarding IBM's future direction and intent are subject to change or withdrawal without notice, and represent goals and objectives only.
- References in this document to IBM products, programs, or services does not imply that IBM intends to make such products, programs or services available in all countries in which IBM operates or does business. Any reference to an IBM Program Product in this document is not intended to state or imply that only that program product may be used. Any functionally equivalent program, that does not infringe IBM's intellectually property rights, may be used instead. It is the user's responsibility to evaluate and verify the operation of any non-IBM product, program or service.
- THE INFORMATION PROVIDED IN THIS DOCUMENT IS DISTRIBUTED "AS IS" WITHOUT ANY WARRANTY, EITHER EXPRESS OR IMPLIED. IBM EXPRESSLY DISCLAIMS ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NONINFRINGEMENT. IBM shall have no responsibility to update this information. IBM products are warranted according to the terms and conditions of the agreements (e.g., IBM Customer Agreement, Statement of Limited Warranty, International Program License Agreement, etc.) under which they are provided. IBM is not responsible for the performance or interoperability of any non-IBM products discussed herein.
- The provision of the information contained herein is not intended to, and does not, grant any right or license under any IBM patents or copyrights. Inquiries regarding patent or copyright licenses should be made, in writing, to:

IBM Director of Licensing
IBM Corporation
North Castle Drive
Armonk, NY 10504-1785
U.S.A.

Trademarks



- The following are trademarks of the International Business Machines Corporation in the United States and/or other countries.

DFDSS
DFHSM
DFSMSdfp
DFSMShsm
DFSMSdss

IBM®
System/390®
SnapShot®
Enterprise Storage
Server®
FlashCopy®
OS/390®
z/OS®

Agenda



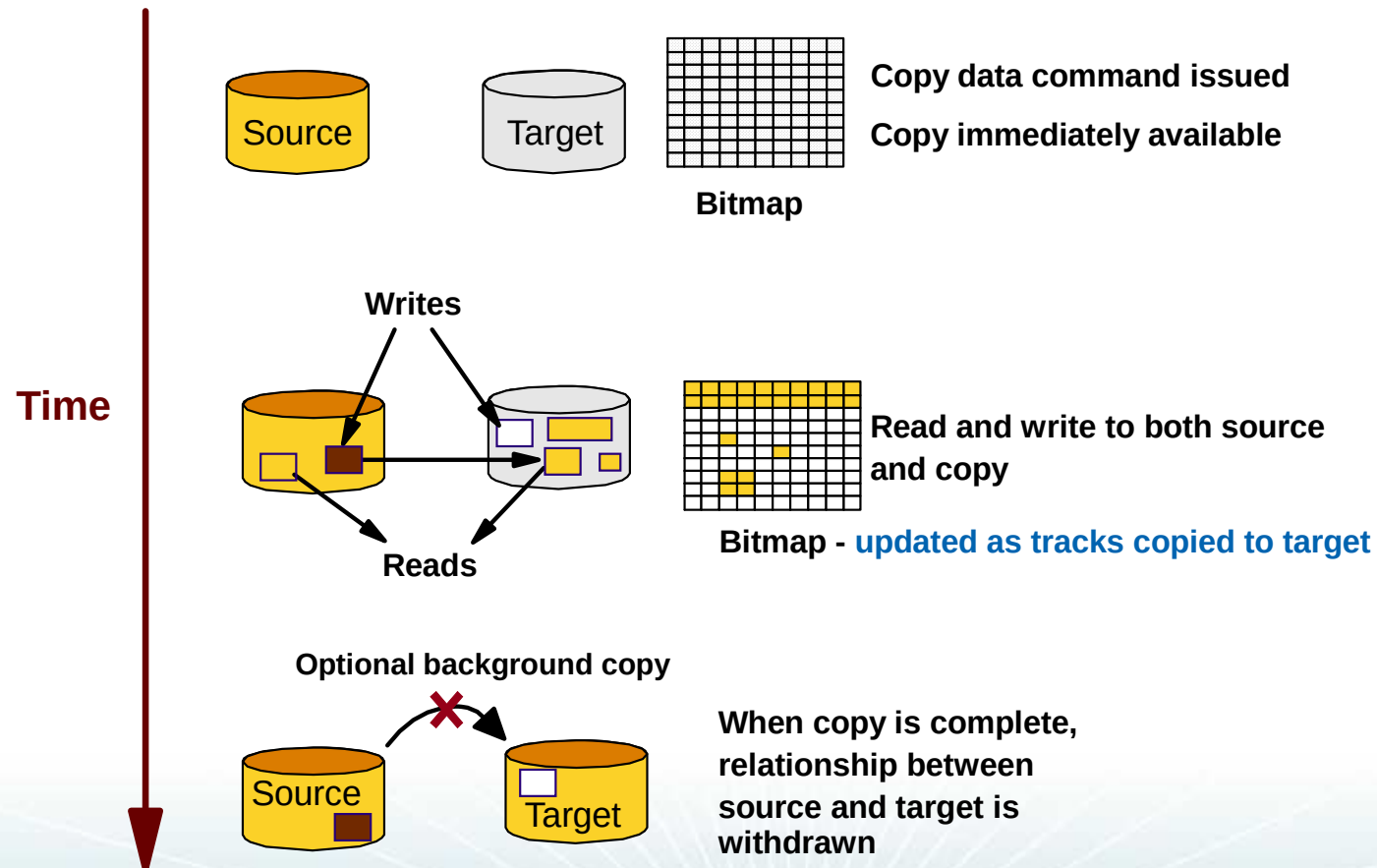
- Introduction to Fast Replication
 - FlashCopy Overview
 - SMS Enhancements
 - DFSMSHsm Enhancements
- Preparing for Fast Replication
- Creating a Backup Copy
- Recovering a Backup Copy
- Withdrawing Background Copies
- Recommendations

Introduction



- DFSMShsm invokes volume-level *fast replication* to create backup versions for sets of volumes
 - Backup hundreds of volumes in a matter of minutes
 - DFSMShsm manages copying the disk copies to tape
 - Developed jointly with DB2
 - System Level Backup utility
- Generic term of "*Fast Replication*"
 - Multiple Point-in-Time copy functions supported by DFSMSdss
 - FlashCopy (Any vendor disk that supports FlashCopy interface)
 - SnapShot
- Recovery can be performed for the entire volume set, individual volumes or individual data sets

FlashCopy Overview



FlashCopy Options

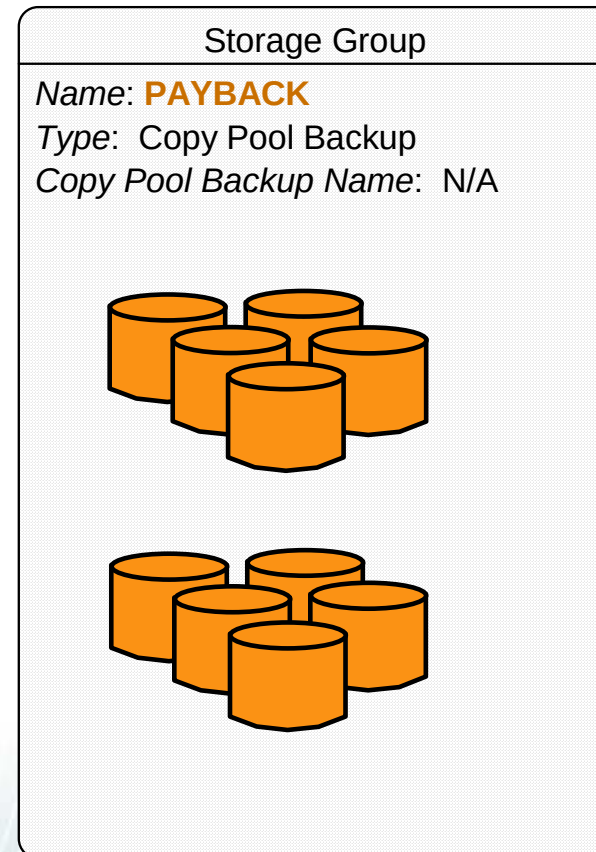
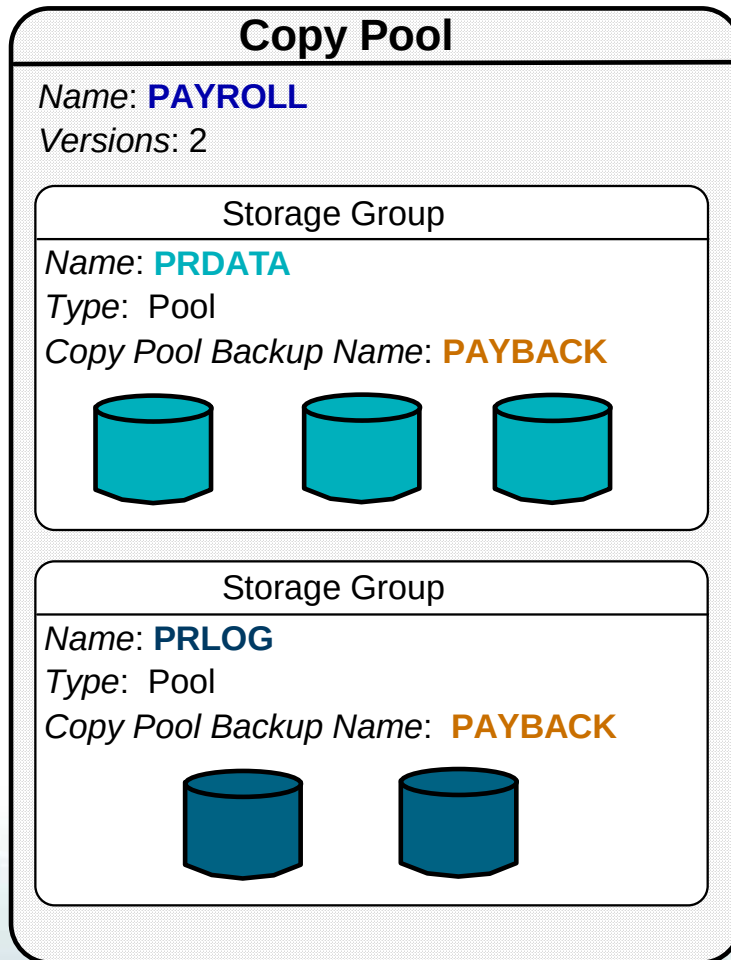
- Full Copy
 - Every track on the source volume is copied to the target volume
- No Copy
 - No background copy is performed
 - Only changed tracks on source are copied to target volume
 - Target volume is used as a cache to create a dump copy
 - *DFSMSHsm does not support Flash back*
- Incremental Copy
 - Initial copy copies every track from source volume to target volume
 - Creates a persistent relationship
 - Only one persistent relationship per source volume is allowed
 - ★ *Subsequent copies only copy tracks that have changed since the last incremental copy*

SMS Enhancements



- New “copy pool” SMS construct
 - Defines which *storage groups* should be processed collectively for fast replication functions
- New “copy pool backup” storage group type
 - Defines which volumes DFSMSHsm may select as target volumes for fast replication backup versions
- ISMF modified to support these SMS enhancements

SMS Enhancements *(continued)*



SMS Enhancements *(continued)*



- New “copy pool” SMS construct
 - Contains from 1 to 256 storage groups
 - Storage groups can be contained within multiple copy pools (Maximum of 50)
 - Specifies the number of *disk* backup copies to maintain
 - 0 – 85 copies
 - ‘0’ indicates that NOCOPY should be used
 - For 1 – 85 copies, a *unique* target volume is required for each source volume!
 - Minimum of 2 versions is recommended (when not using NOCOPY)
 - *Creation of new copy overwrites existing copy*
 - Name: Up to 23 characters
 - Pre-V1R8 was 30 characters. Existing names of 30 characters still honored
- ★ Storage group volumes dynamically retrieved at time of each backup

SMS Enhancements *(continued)*



- New “copy pool backup” storage group type
 - Defines candidate target volumes for DFSMSHsm fast replication
 - Cannot be assigned for allocation by SMS ACS selection routines
 - SMS status ignored by DFSMSHsm
 - Disable New or Disable All does not affect DFSMSHsm selection
 - To prevent DFSMSHsm from using volume, vary it offline
 - New field in Storage Group definition for POOL type storage groups
 - Only applies to POOL type storage groups!
 - For each source volume to be copied in a storage group:
 - There must exist enough eligible target volumes in copy pool backup storage group to satisfy the needs of the # of specified backup versions
 - Eligible target volumes:
 - *Same track format*
 - *Exact size*
 - *Not be in a FlashCopy relationship at the time of selection*
 - *Not in an XRC or PPRC relationship*

DFSMShsm Enhancements



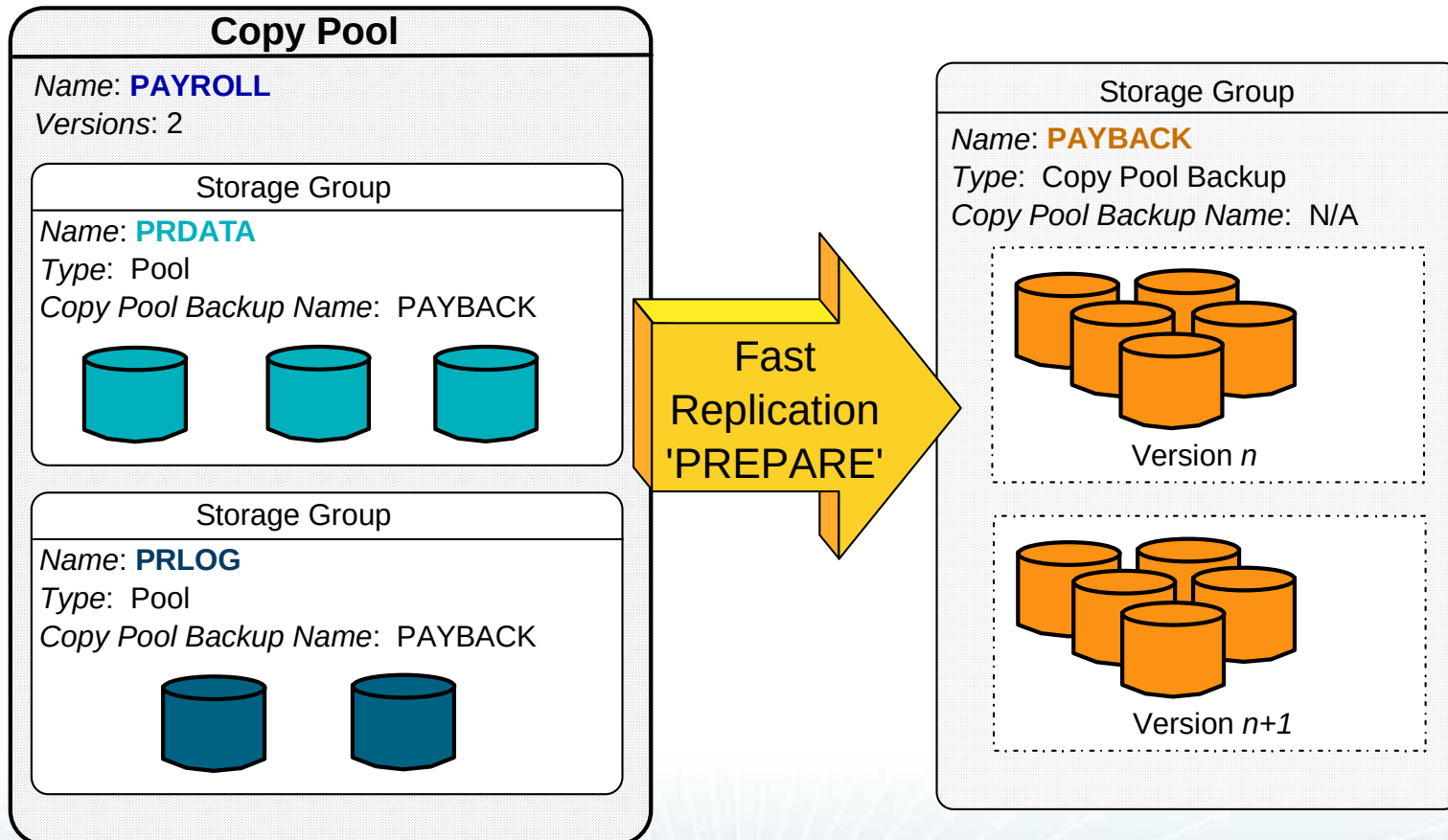
- Three new commands
 - **FRBACKUP** – Create a fast replication backup version for each volume in a specified copy pool.
 - **FRRECOV** – Use fast replication to recover
 - Entire copy pool from a *disk* copy
 - Individual volume from a disk or tape copy
 - One or more data sets from a disk or tape copy
 - **FRDELETE** – Delete one or more unneeded fast replication backup versions
- Modified existing commands
(Discussed in Part 2)
 - **SETYS**, **DEFINE**, **DELVOL** – Define environment
 - **QUERY**, **LIST** and **ARCXTRCT** macro – Monitor versions
 - **HOLD / RELEASE**, **CANCEL** – Manage processing
 - **AUDIT** – *Identify* control data set errors

Preparing for Fast Replication Backup

- **PREPARE** option provided to validate environment and reduce elapsed time of actual backup window
- When PREPARE is specified on FRBACKUP command:
 - For each version > 0, DFSMSHsm pre-assigns a target volume to each source volume in every storage group assigned to the copy pool
 - For NOCOPY (Versions=0), DFSMSHsm only verifies that enough target volumes are available
 - Pairings are maintained in DFSMSHsm control data sets
- The PREPARE function *should be* performed whenever there is a change in the environment, such as:
 - Volumes are added to a storage group
 - # of backup versions changes
 - Storage groups are added to the copy pool
- If PREPARE is not performed, target volume selection occurs during backup window

Preparing for Fast Replication Backup

(continued)

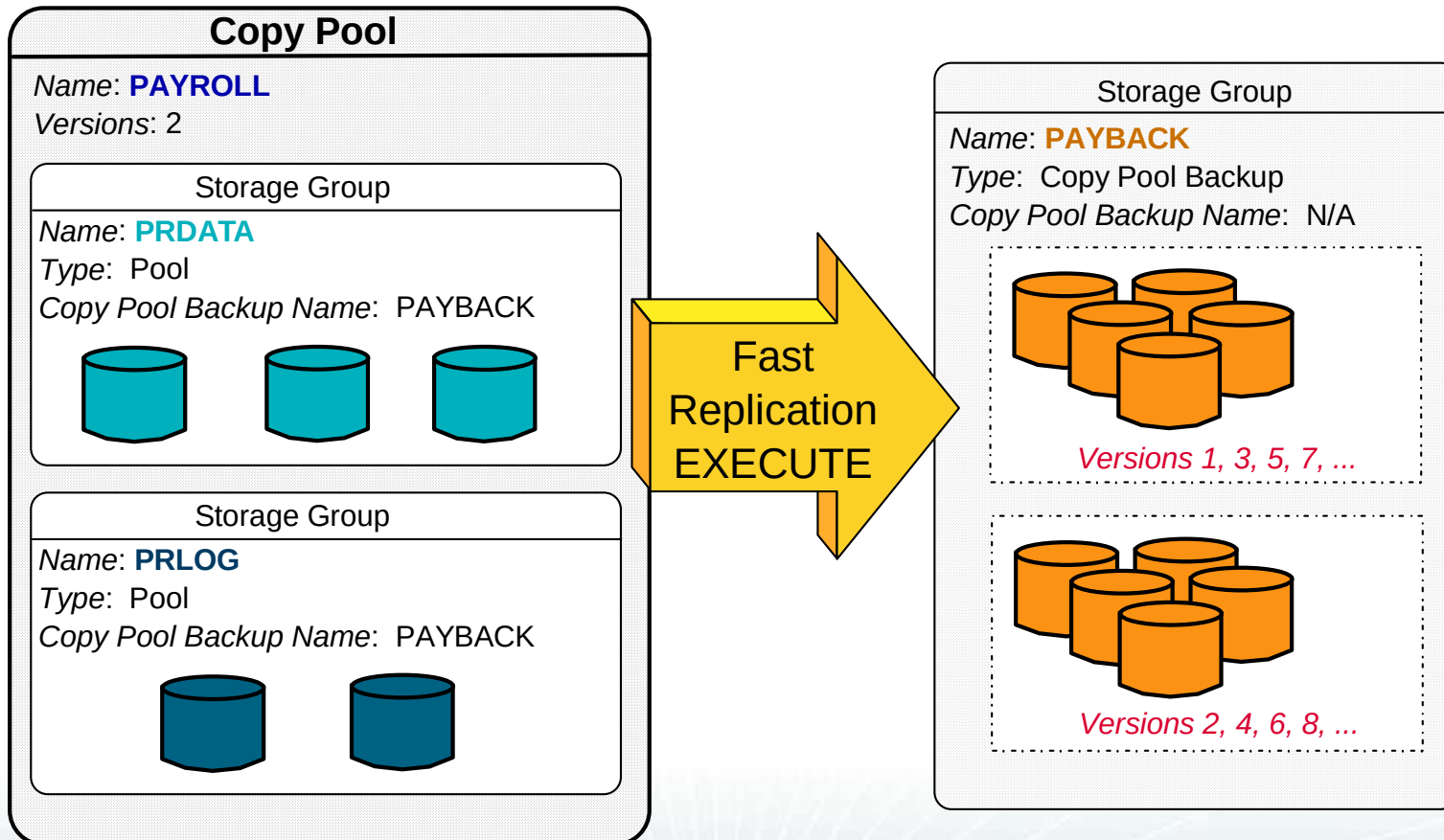


Creating a Fast Replication Backup

- New DFSMShsm command - **FRBACKUP**
 - **FRBACKUP COPYPOOL(cpname) EXECUTE TOKEN(token)**
- Invoked via console, HSEND CMD, batch job or programmatically via the ARCHSEND macro
- Command is considered successful only after a fast replication relationship has been established for every source volume
- If one or more volumes fail:
 - Version is marked as a failure
 - Target volumes that failed are disassociated from source volume
 - All successful relationships are Withdrawn
 - Failed version will be the target of the next FRBACKUP command
 - Version number does NOT increment until successfully created

Creating a Fast Replication Backup

(continued)



Creating a Fast Replication Backup

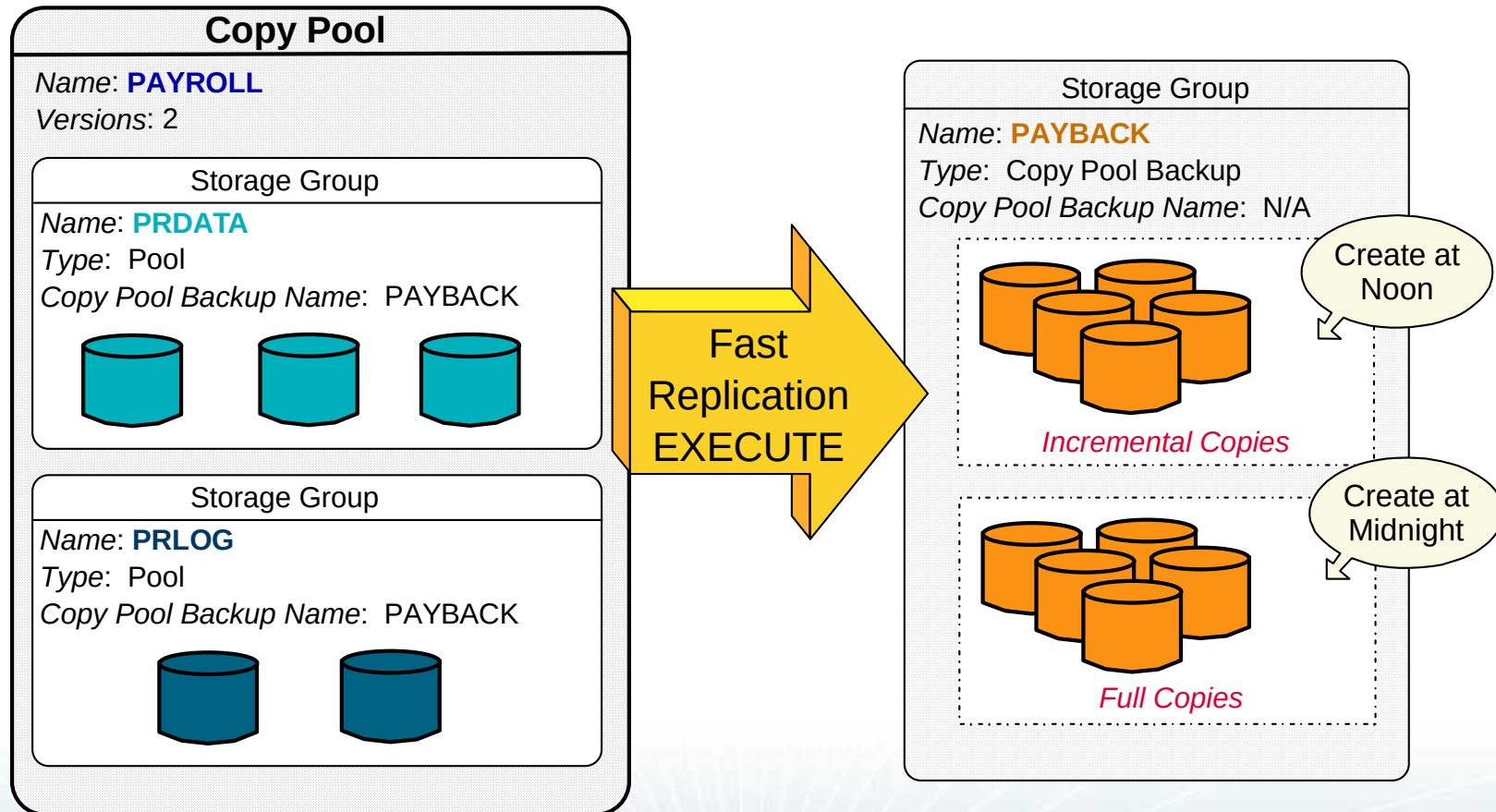
(continued)



- Versions = 1 to 85
 - Default is to use FlashCopy Full for each source volume
 - **FCINCREMENTAL** keyword used to specify a volume set for which FlashCopy Incremental should always be used
 - FCINCREMENTAL only needs to be specified once
 - Each time that volume set is Flashed, an Incremental copy is made
 - **FCINCREMENTALLAST** keyword used to change an Incremental copy to a Full copy
 - Versions = 0
 - NOCOPY option is used to create FlashCopy
 - Source to Target relationship is Withdrawn after the target volume is dumped to tape (Discussed in Part 2)
 - ★ *Target volumes are available to be used by different copy pools*
 - ★ *Remember – Cannot recover from disk copy*
- ✓ FCINCREMENTAL, FCINCREMANTALLAST and VERSIONS=0 are only available in V1R8

Creating a Fast Replication Backup

(continued)



Creating a Fast Replication Backup

(continued)



- The TOKEN parameter:
 - User specified identifier for a particular backup version
 - Maximum 40 characters
 - Use to identify a version for Recovery or Deletion
- Serialization
 - Individual volume VTOCs are serialized as they are flashed
 - Responsibility of caller to protect source volume against data set level updates until FlashCopy relationship established
 - If all volume VTOCs need to be serialized concurrently, caller must do so
 - **NOVTOCENQ** keyword indicates that VTOCs have already been serialized

Fast Replication Backup Parallelism

- FRBACKUP COPYPOOL commands are processed in serial, however:
 - Volumes in copy pool are processed in parallel
 - DFSMSdss invoked with PARALLEL option and up to 254 volume pairs
 - DFSMShsm will process between 1 and 64 DFSMSdss invocations concurrently
 - SETSYS MAXCOPYPOOLTASKS(FRBACKUP(*nn*) DSS(*mm*))
 - FRBACKUP(*nn*) is # of concurrent DFSMSdss invocations
 - Default is **15**
 - DSS(*mm*) is # of volume pairs that DFSMShsm will pass to each DFSMSdss invocation
 - Default is **24**
- ★ Defaults represent recommended values for optimal performance based on internal testing

Recovering a Fast Replication Backup

- Target volumes cannot be used as a source volume
 - Dump conditioned volumes
- DFSMSHsm **FRRECOV** command must be used to recover target volume back to the source
- **FRRECOV COPYPOOL(cpname) VERIFY(Y)**
 - Recovers all of the volumes from the named copy pool
 - Retry failed volumes individually with TOVOLUME option
- **FRRECOV TOVOLUME(volser)**
 - Recovers an individual volume
- **FRRECOV DSNNAME(dsname1, dsname2, ...) REPLACE**
 - Recovers one or more data sets

✓ DSNNAME is only available in V1R8

Data Set Recovery



! Restriction

- At the time of the recovery, data sets *must be* cataloged on the same volumes that they resided on at the time of the backup
- No support for:
 - Deleted data sets
 - Data sets that have moved since the time of the backup
- Support for:
 - Data sets that have extended to new volume(s) since the backup
- Data set names may be fully or partially qualified
 - '**' not allowed as first or only qualifier
 - Standard qualifiers are accepted
 - Example: USER*. **
 - Every data set in the catalog standard order of search will be processed
- Multiple data sets on the same command are processed individually
 - After all data sets have been processed, completion messages are issued

Data Set Recovery *(continued)*



- **Unsupported data set types**
 - Catalogs
 - VVDS
 - VTOC Index
 - VSAM Key range
 - Migrated data sets
- **VSAM data sets**
 - Processed at the cluster level only
 - AIxes are processed independently of the base cluster

Recovering a Fast Replication Backup

(continued)



- If recovering other than the most recent version, use keywords:
 - GENERATION
 - VERSION
 - TOKEN
 - DATE
- Cannot recover at Copy Pool or Volume level from disk until the Background Copy (BGC) has completed
 - QUERY COPYPOOL(cpname) - Lists which volumes are in an active relationship and the BGC percent complete
 - *(Percent complete is only shown in V1R8)*
 - Before recovery begins at the copy pool level, all volumes are queried to determine if BGCs are complete
 - Do Not override with VERIFY(N) keyword unless using SNAP/SHOT
- Data set recovery can begin before BGC is complete
 - Traditional I/O is used

Recovering a Fast Replication Backup

(continued)



- Data Set Recovery I/O Technique selection
 - Fast Replication options
 - **REQUIRED** – Only recover if fast replication can be used
 - **PREFERRED** – If fast replication can't be used, use traditional I/O
 - *Fast replication cannot be used if BGC has not completed*
 - **NONE** – Only recover using traditional I/O. Data set cannot be used until I/O is complete. (*Much slower*)
- SETSYS FASTREPLICATION(DATASETRECOVERY(...))
 - Sets the technique to use
 - Default is PREFERRED
- Optional FASTREPLICATION(REQ|PREF|NONE) keyword on FRRECOV command overrides SETSYS value

Fast Replication Recover Parallelism

- FRRECOV COPYPOOL commands are processed in serial, however:
- Volumes in copy pool are processed in parallel
 - DFSMSdss invoked with PARALLEL option and up to 254 volume pairs
 - DFSMShsm will process between 1 and 64 DFSMSdss invocations concurrently
- SETSYS MAXCOPYPOOLTASKS(FRRECOV(*nn*) DSS(*mm*))
 - FRBACKUP(*nn*) is # of concurrent DFSMSdss invocations
 - Default is **15**
 - DSS(*mm*) is # of volume pairs that DFSMShsm will pass to each DFSMSdss invocation
 - Default is **24**

Withdrawing a DFSMShsm Fast Replication Backup Version



- ! NEVER Withdraw FlashCopy relationships outside of DFSMShsm control!
 - Will cause backup version to be invalidated, but DFSMShsm will process it as valid
 - Could result in data integrity exposure
- Use the FRBACKUP COPYPOOL(cpname) WITHDRAW command
 - Will withdraw all current FlashCopy relationships
 - Applies to most recent VALID version of the specified copy pool
 - The version state will be changed to FAILED
- Can only be performed at the copy pool level

Recommendations

- Define catalog within copy pool
 - Otherwise, recovery will cause catalog and data to be out-of-synch
 - Unallocate all catalogs in the copy pool before recovery
- **DO NOT Migrate data within copy pool**
 - Fast Replication is a track-level operation
 - Migrating / Recalling data within a copy pool will cause data loss during recovery
 - Recalling a data set in-between a backup and a recovery will cause the recalled data set to be overwritten
 - This same restriction exists for physical operations today
- Patch to determine why a target volume is not being selected
 - Warning – Lots of messages!
 - PATCH .FRGCB.+9 BITS(.1.....)
- BCDS record size *must* be 6544